

MATERIAL SAFETY DATA SHEET

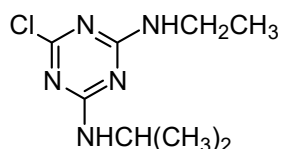
1. Chemical Product Identification

Product Name: ATRAZINE 90% WG

Molecular Formula: $C_8H_{14}ClN_5$

Molecular weight: 215.7

Structural formula:



Chemical Name: 2-chloro-4-ethylamine-6-isopropylamino-S-triazine

CAS No.: 1912-24-9

Chemical group: 1, 3, 5-triazine Herbicide

Manufacturer: Shandong Binnong Technology Co., Ltd.

Address: No.518, Yongxin Road, Binbei Town, Binzhou, Shandong, China

2. Composition / Information on Ingredients

Ingredient		CAS No.	Content % (w/w)
Active ingredient	Atrazine	1912-24-9	90.0
Inert ingredient	Other ingredients, including water	//	Up to 100%

3. Hazards Identification

Toxicity class: WHO Table 5; EPA III

Main hazard: Eye irritant. (Reversible)

Flammability: Not flammable.

Biological hazards:

Eye contact: May cause pain, redness or tears.

Skin contact: Minimally toxic and practically non-irritating.

Ingestion: Moderately toxic. No significant effects are expected to develop if only small amounts are swallowed.

Inhalation: Minimally toxic if inhaled.
Carcinogenicity: See section 11.
Mutagenicity: See section 11.
Reproductive/Teratogenicity: See section 11.

4. First Aid Measures

The acute toxicity to triazines for man is thought to be low, and no adverse health effects from exposure to this herbicide have been reported. In view of the toxicity induced in experimental animals on repeated exposure, proper care should be taken during occupational use to avoid excessive inhalation of dust or spray particles, and to prevent accidental contamination of food products and water.

Inhalation:

Remove source of contamination or move victim to fresh air. Obtain medical advice immediately.

Skin contact:

If irritation occurs, remove contaminated clothing, shoes and leather goods (e.g. watchbands, belts). Gently wipe off excess chemical. Wash skin gently and thoroughly with water and non-abrasive soap. If irritation persists, seek medical advice immediately. Persons who become sensitized may require specialized medical management with anti-inflammatory agents.

Eye contact:

Immediately flush the contaminated eyes with gently flowing lukewarm water for 20 minutes, holding the eyelid(s) open. Severe contamination may require medical attention.

Ingestion:

Have victim rinse mouth thoroughly with water. Do not induce vomiting. In serious cases, seek medical advice immediately.

Advice to physician:

No signs and symptoms of triazine poisoning are known or expected in humans. An antidote is neither known nor needed. Treat symptomatically when required. Induction of emesis is not recommended because there is so little information about the effects of overdose in humans. When large amounts have been ingested, gastric

lavage or the administration of activated charcoal with water may be indicated.

5. Fire-Fighting Measures

Extinguishing agents:

Extinguish small fires with carbon dioxide, dry powder, or alcohol-resistant foam. Water spray can be used for larger fires or cooling of unaffected stock, but avoid the accumulation of polluted run-off from the site. Remove container from fire area if possible. Contain fire control water for later disposal. Use a recommended extinguishing agent for the type of surrounding fire. Avoid inhaling hazardous vapors. Keep upwind.

Special hazards:

The substance decomposes on heating producing toxic fumes including hydrogen chloride and nitrogen oxides.

Personal protective equipment:

Fire fighters and others that may be exposed should wear full protective clothing and self-contained breathing apparatus.

6. Accidental Release Measures

Personal precautions:

Avoid contact with skin and eyes. Do not breathe in fumes. For personal protection see Section 8.

Environmental precautions:

Do not allow entering drains or watercourses. When the product contaminates public waters, inform appropriate authorities in accordance with local regulations.

Small dry spills, sweep up with damp earth or sand or other suitable absorbent, such as sawdust, taking care not to raise a dust cloud. Place the material into a clean, dry container and cover for subsequent disposal.

Large spills; evacuate area. Wear appropriate protective clothing. Ventilate area, sweep up and place in an appropriate container. Wash contaminated surfaces to remove any residues. In situations where product comes in contact with water, contain contaminated water for later disposal. Do not flush spilled material into drains. Keep unnecessary people away.

7. Handling and Storage

Handling:

Harmful by inhalation, if swallowed and contact with eyes and skin. Avoid contact with eyes, prolonged contact with skin, and inhalation of dust. Use with adequate ventilation. Wash hands before eating, drinking, chewing gum, smoking, or using the toilet. Remove clothing immediately if the herbicide gets inside, and then wash skin thoroughly using a non-abrasive soap and put on clean clothing. Do not apply directly to areas where surface water is present, or to intertidal areas below the mean high water mark. Water used to clean equipment must be disposed of correctly to avoid contamination.

Storage:

Store in its original labeled container in shaded, well-ventilated area, away from heat, sparks and other sources of ignition. Not to be stored next to foodstuffs and water supplies. Keep out of reach of children and animals. Local regulations should be complied with.

8. Exposure Controls/Personal Protection

PROTECTION

Occupational exposure limits:

TLV: ppm; 5 mg/m³ (ACGIH)

Engineering control measures:

If engineering controls and work practices are not effective in controlling exposure to this material, then wear suitable personal protective equipment including approved respiratory protection.

Personal protective equipment:

It is essential to provide adequate ventilation. The measures appropriate for a particular worksite depend on how this material is used and on the extent of exposure. Ensure that control systems are properly designed and maintained. Comply with occupational safety, environmental, fire, and other applicable regulations.

Respirator:

An approved respirator suitable for protection from dusts and mists of pesticides is

adequate. Limitations of respirator use specified by the approving agency and the manufacturer must be observed.

Clothing:

Employee must wear appropriate protective (impervious) clothing and equipment to prevent repeated or prolonged skin contact with this substance.

Gloves:

Employee must wear appropriate synthetic protective gloves to prevent contact with this substance.

Eye protection:

The use of goggles is recommended. Emergency eye wash: Where there is any possibility that an employee's eyes may be exposed to this substance; the employer should provide an eye wash fountain or appropriate alternative within the immediate work area for emergency use.

9. Physical and Chemical Properties

Appearance:	White granules.
Odor:	Odorless.
Explosive properties:	Non explosive.
Oxidizing properties:	Not oxidative.
PH:	No data available.
Relative density:	Average density is $1.240 \pm 0.002 \text{ g/cm}^3$ at 25.5°C .
Storage stability:	Atrazine was determined to be stable in the dry state at low and elevated temperatures.
Persistent foaming:	Not applicable as the product is a solid.
Suspensibility:	No data available.
Dilution stability:	No data available.
Solubility in water:	33 mg/ at 20°C .
Partition-coefficient in n-octanol / water:	$\log P_{o/w} : 2.54 (25^\circ\text{C})$
Flash point:	Not applicable as product is a solid at ambient temperature.

10. Stability and Reactivity

Stability:

Only slightly sensitive to natural light and extreme temperatures. Relatively stable in neutral, weakly acidic and weakly alkaline media, but rapidly hydrolysed to the hydroxy derivative in strong acids and alkalis, and at 70°C in neutral media. The product has a shelf life of 2 years, provided it is stored in its unopened, undamaged original containers in well-ventilated and dry conditions away from sources of heat and spark generating equipment.

Incompatibility:

Spray solutions containing this product should be mixed, stored or applied using stainless steel, aluminium, fibreglass or plastic-lined containers. Product is relatively stable in neutral, weakly acidic and weakly alkaline media. The product is compatible with most other pesticides and fertilisers when used at normal rates. However, a compatibility test is required before using with other products. Do not physically mix concentrate directly with other herbicides or pesticide concentrates; always dilute first. The product may flocculate in the presence of paraquat.

Thermal decomposition:

Hydrogen chloride and toxic oxides of nitrogen are released when the product decomposes on heating.

11. Toxicological Information

Acute oral LD₅₀:	3125 mg/kg in male rats. 4883 mg/kg in female rats
Acute dermal LD₅₀:	2000 mg/kg in rats (male and female).
Acute inhalation LC₅₀:	LC ₅₀ (4h) for rats (male and female): 5.8 mg/ of air.
Acute skin irritation:	The test substance was found to be non-irritating to rabbit skin and not corrosive after exposure for 4 hours.
Acute eye irritation:	Atrazine is considered to be a moderate irritant to the ocular tissue of the rabbit. The product does not cause any permanent eye damage, or any severe or lasting irritation or corrosion.
Dermal sensitisation:	Atrazine does not have any dermal sensitisation properties.
Carcinogenicity:	Long-term animal studies did not show carcinogenic activity.

No human information is available. ACGIH: A4

Teratogenicity: Animal studies did not produce any effects. No human information available.

Mutagenicity: Not mutagenic in a series of tests using bacteria, cultured mammalian cells and whole animals. No human information available.

12. Ecological Information

Ecotoxicology:

Birds: Very low toxicity.

Acute oral LC₅₀: 5000 mg/kg (bobwhite quail, Japanese quail, ring necked pheasant and mallard duck)

Acute oral LC₅₀: 10 000 mg/kg (peking ducks)

Fish: May pose a hazard to fish.

Acute toxicity for several fish species (rainbow trout, carp, bluegill, and guppies):

the 96 hour LC₅₀ varied between 4.3 mg/ λ (guppies) and 100 mg/ λ (carp).

Bees: Relatively non-toxic.

LD₅₀ (oral): > 97 μ g / bee.

Daphnia: May pose a hazard to Daphnia.

EC₅₀ (48 hr) : 6.9 mg/ λ .

EC₅₀ (21 d) : 0.12 mg/ λ .

Algae (72 h): Toxic to algae.

Scenedesmus subspicatus: 0.043 mg/l (Tech)

Selenastrum capricornutum; 0.13 mg/l (tech)

Earthworms: Low toxicity for earthworms.

Soil micro-organisms:

With the exception of transient inhibition of bacterial growth during the first week after product application, repeated applications of the product did not affect the number of viable bacteria or fungi, relative abundance of bacteria producing hydrolytic enzymes or soil enzyme levels.

Degradability:

Strongly adsorbed to soil. Microbial degradation is the major cause of loss from soil. Half-life in soil is between 35 and 50 days, but may be longer under cold or dry conditions. The half-life under groundwater conditions is 105 to 200 days. The main metabolites under all conditions are desethylatrazine and hydroxyatrazine.

Mobility:

The product is relatively mobile in sandy soils and can result in the contamination of surface and ground water.

Accumulation:

The product shows little or no tendency to bioaccumulate and poses no long term threat to wildlife.

German wgk: 2

13. Disposal Considerations

Pesticide disposal:

Contaminated absorbents, used containers, surplus product, etc., should be burnt in an incinerator, preferably designed for pesticide disposal. Hydrolysis under alkaline conditions (10% w/v sodium hydroxide) is a suitable method to dispose of small quantities of atrazine. Heating speeds up the process. After hydrolysis, dilute and dispose of via the sewage system. Atrazine is relatively stable and characterised by high mobility in some soils and should not be buried in dump sites, landfills, etc. Comply with any local legislation applying to waste disposal.

Package product wastes:

Emptied containers retain vapor and product residues. Observe all labeled safeguards until container is cleaned, reconditioned, or destroyed. Combustible containers should be disposed of in pesticide incinerators. Non-combustible containers must first be triple-rinsed with water, punctured and transported to a scrap metal facility for recycling or disposal.

14. Transport Information

UN Number:	3077
Class:	9
Packing Group:	III
Substance:	Environmentally hazardous substance, solid, n.o.s

(Atrazine 900 g/kg)

15. Regulatory Information

Symbol: Xn , Xi
EU Classification: Carc. Cat. 3; Mut. Cat. 3
Indication of danger: Harmful; Irritant.

Risk phrases:

R20/22 Harmful by inhalation and if swallowed.
R 36 Irritating to eyes.
R40 Possible risks of irreversible effects.
R43 May cause sensitization by skin contact.

Safety phrases:

S 2 Keep out of reach children.
S 36/37 Wear suitable protective clothing and gloves.
S 46 If swallowed seek medical advice immediately and show this container and label.

National Legislation:

In accordance with the South African National Road Traffic Act, 1996(Act 93 of 1996), the Fire Brigade Act, 1987(Act 99 of 1987) and the Occupational Health and Safety Act, 1993 (Act. No. 85 of 1993).

16. Other Information

All information and instructions provided in this Material Safety Data Sheet (MSDS) are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be correct. This information applies to the PRODUCT AS SUCH. In case of formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons in receipt of this MSDS to ensure that the information contained herein is properly read and understood by all people who use, handle, dispose or in any way come in contact with the product. If the recipient subsequently produces formulations(s) containing this product, it is the recipient's sole

responsibility to ensure the transfer of all relevant information from this MSDS to their own MSDS.